

openQSE Day Quantum Resource Interface Workstream

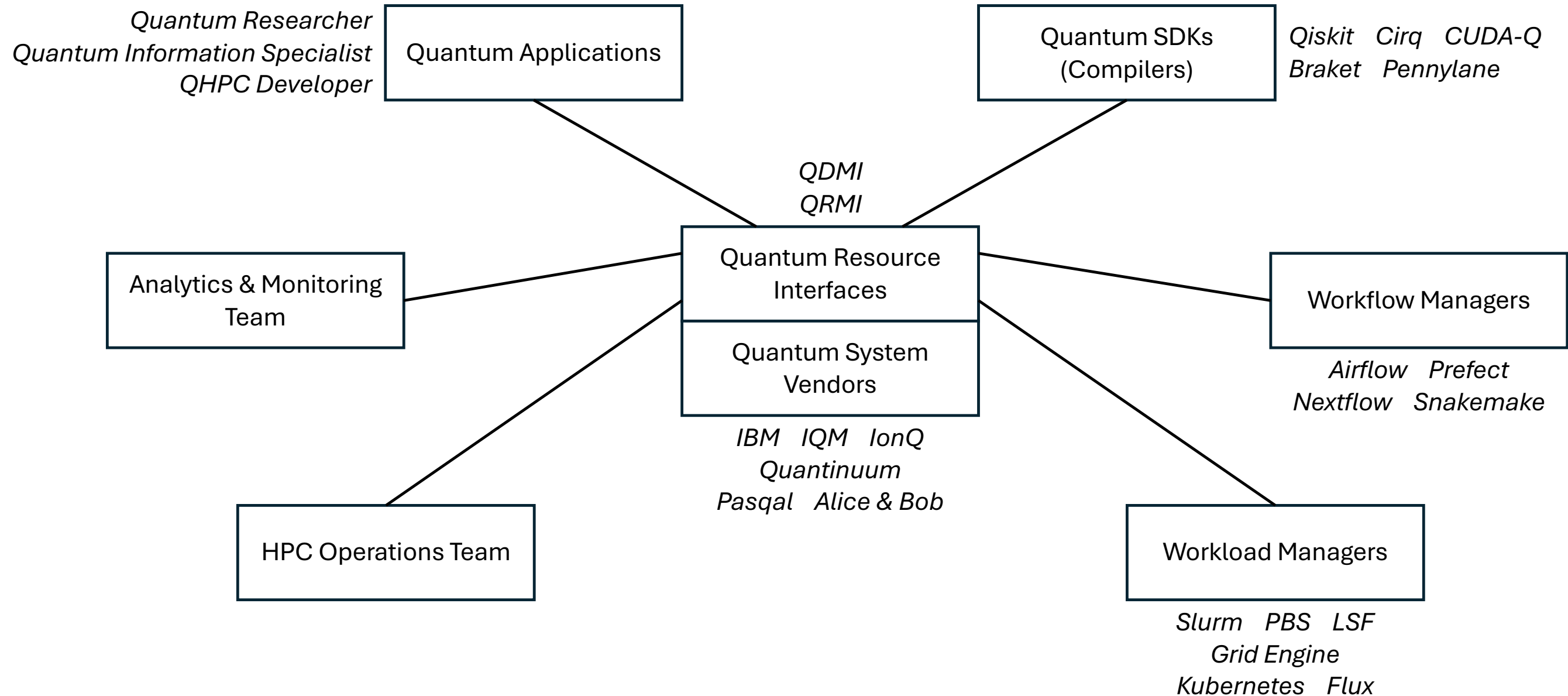
Organizers: Yoon Park & Thomas Naughton

24 July 2026

Overview

- Thank you to Amir Shehata and ORNL leadership!
- Goals
 - *Ideal: Establish a standard, community-driven resource management interface to integrate quantum systems into HPC data centers*
 - Realistic: Define a methodology – rinse and repeat the following?
 - Document requirements from stakeholders
 - Interface developers implement APIs to “meet” requirements
 - Run experiments to evaluate
 - Today: Create a list of milestones, publications, artifacts, etc

Stakeholders (You)



Format

- Lightning talks
 - Level set on nomenclature
 - What is the current state of quantum resource interfaces?
 - QDMI, QRMI, QFw
 - What are the requirements from stakeholders?
 - Workload managers
 - Other workstreams
 - HPC operations team
 - Analytics and monitoring team
- Each speaker will leave ample time for discussion
- EOD: Create a list of milestones, publications, artifacts, etc

Quantum Resource Interface Workstream

10:15 AM	12:30 PM	135	Workstreams - Session 1	10:15 AM	10:20 AM	5	Overview - Yoon Park
				10:20 AM	10:30 AM	10	Nomenclature - Munetaka Ohtani
				10:30 AM	11:00 AM	30	QDMI - Matthias Traube
				11:00 AM	11:30 AM	30	QRMI - Munetaka Ohtani
				11:30 AM	12:00 PM	30	QFw - Doug Oucharek
				12:00 PM	12:30 PM	30	Slurm - Doug Jacobsen
12:30 PM	1:00 PM	30	Working Lunch				
1:00 PM	1:20 PM	20	Group Photo				
1:20 PM	2:00 PM	40	Keynote - Laura Schulz				
2:00 PM	2:15 PM	15	Break				
2:15 PM	4:15 PM	120	Workstreams - Session 2	2:15 PM	2:35 PM	20	Flux and Scheduling - Vanessa Sochat
				2:35 PM	2:55 PM	20	PBS - Cameron Brunner
				2:55 PM	3:15 PM	20	LSF - Gabor Samu
				3:15 PM	3:35 PM	20	Sys arch & workflows, Compiler work streams
				3:35 PM	3:55 PM	20	ORNL HPC operations team - Michael Sandoval
				3:55 PM	4:15 PM	20	ORNL analytics & monitoring team - Tim Osborne
4:15 PM	5:30 PM	75	Readouts				

Sessions – 0

Session	Speaker	Takeaways
Nomenclature	Munetaka Ohtani (IBM)	Focused on how quantum systems operate and interact within HPC environments
QDMI	Matthias Taube (Munich)	Will focus on modularity, eg, core interface + extension modules
QRMI	Munetaka Ohtani (IBM)	Will focus on workload manager integration, eg, resource discovery, accounting information, credentials management
QFw	Doug Oucharek, ORNL	QDMI/QRMI comparison almost ready to start, adding transport, benchmarking, scheduler <i>Can we add Flux to QFw testbed?</i>

Sessions – 1

Session	Speaker	Takeaways
Slurm	Doug Jacobsen (NVIDIA)	• SPANK plugins are increasingly viewed as a maintenance and support burden • Recommendation – Consider using HRES • HRES will replace dynamic licenses while adding plugin-based state validation • HRES currently relies on Lua • Claim – Slurm operators favor Lua because scripts are plain-text and operationally approachable • Workstream’s concern – Practical implementations often require external packages, reducing some of the simplicity benefits commonly associated with Lua

Session – 2

Session	Speaker	Takeaways
Flux	Vanessa Sochat (LLNL)	• Demonstration of a solution to the "two-queue problem" using Fluxion and Kubernetes • Work is transitioning toward Flux Core • Demonstrated QRFI integration with Snakemake workflow tool
PBS	Cameron Brunner (Siemens)	• QPU pooling concept • Scalability concerns associated with runJob-based approaches • Discussion of alternative approaches, although time constraints prevented deeper exploration • PBS will offer evaluation licenses for the full product suite to QRFI community

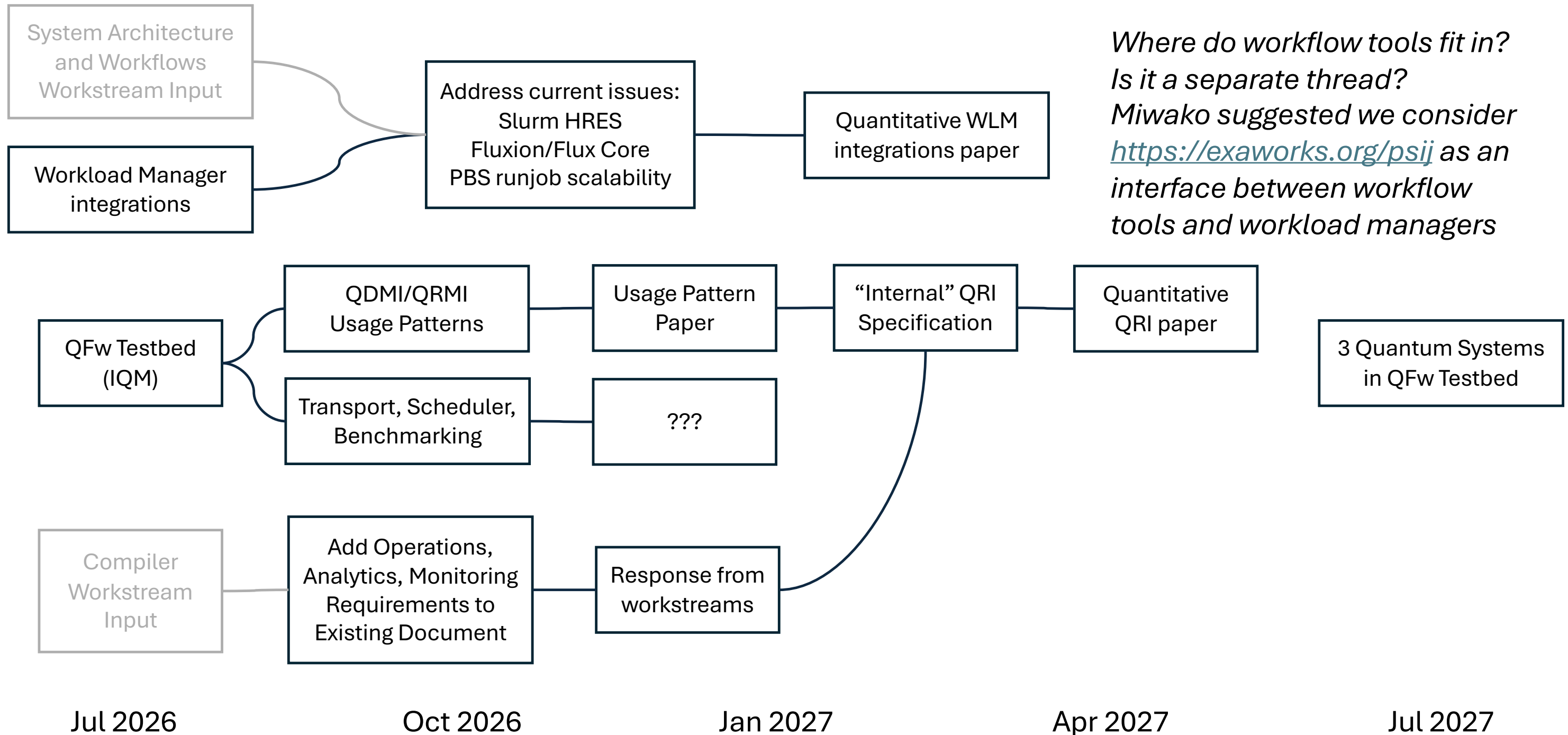
Sessions – 3

Session	Speaker	Takeaways
LSF	Gabor Samu (IBM)	• Demonstrate SQD workflow • Provided resource discovery via IBM-specific API • Wish list • Accounting information – What was consumed, what was the efficiency, how much did it cost? • Task pending times – Estimate of queue depth and pending time for tasks
ORNL Operations	Michael Sandoval, (ORNL)	MANY issues to address for operators and users <i>Ryan will share a paper that describes issues</i>

Milestones

Date	Type	Description	Owners
Mar/Apr 2026	Paper	Standardizing Access to Heterogeneous Quantum Backends: A Case Study on Cloud Service Integration with QDMI (https://arxiv.org/abs/2603.05138) Practical HPCQC Integration with QDMI: A Real-Hardware Case Study with IQM Systems (https://arxiv.org/abs/2604.19869)	QDMI community
Jul 2026	Paper	Hybrid Quantum and Classical Workload Management with Graph-based Scheduling (https://arxiv.org/abs/2607.09151)	Vanessa S, Dan M
Jul 2026	Paper	Examination of QRMI as a unified interface for quantum-HPC workloads (https://arxiv.org/abs/2607.19591)	QRMI community
Aug 2026	Meeting series	Establish quantum resource interface meetings	Yoon P
Aug 2027	Paper	Quantum resource interface requirements for operations and monitoring	Michael S, Tim O
Sep 2026	Testbed	QFw IQM testbed Do we split this in usage patterns, transport, benchmarking, admissions control/scheduler?	Doug O, Amir S
Oct 2026	Paper	Response to interface requirements for operations and monitoring – NOT implementation	Matthias T, Munetaka O, Doug O
Dec 2026	Data	Collect QDMI/QRMI usage pattern data on QFw testbed	Doug O, Amir S
Jan 2027	Paper	Usage patterns of QDMI/QRMI on QFw testbed – collaborate with Ryan L on computational patterns	Matthias T, Munetaka O
Jan 2027	Testbed	Evaluate Slurm HRES with QDMI/QRMI – can we produce something sooner	Matthias T, Munetaka O, Doug J, Stefano M
Feb 2027	Specification	First openQSE quantum resource interface specification (DRAFT)	Doug O, Amir S
Mar 2027	Data, Paper	Quantitative analysis of QDMI/QRMI as a unified interface for quantum-HPC workloads	Matthias T, Munetaka O
		Evaluate PBS QPU pooling concept	Cameron
		Add other workload managers to QFw testbed – which ones? Flux? PBS? LSF? Grid Engine?	???
		Add workflow tools to QFw testbed, establish workflows on QFw testbed	???
Jul 2027	Testbed	Add 2 additional systems to QFw testbed (total 3 by the end of year 1)	Doug O, Amir S

Workstream Threads



Notes

- 20 attendees
- “Cost” is really “metering”
- Add “status/health” to quantum resource definition
 - Do users require notifications about “status/health”?
 - Should the workload manager re-run jobs automatically?
- Telemetry (eg, machine state information) is not needed for scheduling (Slurm)
 - Just need usage for scheduling
- Is IQM storage configurable? 2 nodes dedicated for this
- QDMI WIP – leaner modularity through core interface + extension modules (eg, provider module + QPU module)
- MQSS benchmarking framework – resources in Teams chat
- Different modalities make benchmarking challenging
- Where does slurmd run?